

IISc Grounded Agentic RAG Platform

M6 n8n Retrieval Module — Live Evaluation Report

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Branch: `codex/evaluation` | Dataset: 148 cases / 13 applications | Judge: RAGAS 0.1.10 (OpenAI gpt-4o-mini, independent)
System under test: `n8n-workflows/retrieval-pipeline.json` running live via FastAPI `/chat/query` → n8n → Neon pgvector → LLM

1. Executive Summary

A real end-to-end evaluation was run against the live n8n retrieval pipeline: `run_eval` → FastAPI `/chat/query` → n8n webhook → Neon pgvector (Voyage `voyage-4-large` embeddings) → generation (`deepseek-v4-flash` with Gemini self-check + OpenAI fallback). Answers were graded by an independent OpenAI judge via RAGAS. **145 of 148 cases scored** (3 skipped on transient upstream 502s).

FAITHFULNESS 0.86 thr 0.85 ✓	ANSWER RELEVANCY 0.67 thr 0.80 ✗	CONTEXT PRECISION 0.93 thr 0.75 ✓	CONTEXT RECALL 1.00 thr 0.80 ✓
CITATION COVERAGE 1.00	NEGATIVE ABSTENTION 1.00	LATENCY MEAN 7.3s	LATENCY P95 / MAX 21/36s

Metric	Score	Threshold	Gate
Faithfulness	0.860	0.85	PASS
Answer relevancy	0.667	0.80	FAIL
Context precision	0.934	0.75	PASS
Context recall	1.000	0.80	PASS

Overall quality gate: FAILED — solely because answer relevancy (0.67) is below 0.80. Retrieval is excellent (recall 1.00, precision 0.93), grounding is solid (faithfulness 0.86), and abstention on unanswerable questions is perfect (1.00). The weak spot is answer *relevancy*: answers are correct and grounded but too terse (e.g. "15 minutes."), which the RAGAS relevancy judge penalizes.

2. Fixes Applied to Make the Evaluation Possible

The system could not be evaluated out-of-the-box. Fixes fall into three groups. **The scores in this report are for the M6 workflow with group A applied**; the committed `retrieval-pipeline.json` still contains these defects.

A. M6 retrieval workflow defects (functional blockers)

- A1. Question field mismatch (blocker).** Embed Query, Rerank Results, DeepSeek Generate and Gemini Self-Check read `${'Webhook'}.item.json.body.query`, but the backend sends `current_message`. Result: empty query → the model received "Question: undefined" and abstained on every case. **Fix:** read `body.current_message` (fallback `body.query`).
- A2. No tenant filter (security + correctness).** Vector Search ran `SELECT ... FROM document_chunks ORDER BY embedding <=> ... LIMIT 10` with no `WHERE tenant_id`, and `tenant_id` was never read — cross-tenant data leakage and polluted retrieval. **Fix:** add `WHERE tenant_id = '{{body.tenant_id}}':::uuid`.
- A3. No sources returned (blocker).** Respond to Webhook returned only `{answer, model_used}`; the backend reads `response.sources`, so citations were always empty and RAGAS context metrics would be ≈0. **Fix:** return `sources` = Vector Search rows mapped to `{chunk_text: content, title: metadata.title}`.
- A4. Invalid generation model.** Node used `deepseek-chat`, which is not in this account's model list (only `deepseek-v4-flash` / `deepseek-v4-pro`). **Fix:** `deepseek-v4-flash`.

B. Environment & dependency fixes (to run the stack with no Docker)

- B1. n8n native build.** System Node v26 is too new for n8n's `isolated-vm` native module. Installed `brew node@22`; node-gyp also needed `distutils` (removed from Python 3.12+) — created a `python@3.12` venv with `setuptools` and passed it via `npm_config_python`. Then n8n

2.26.8 built.

B2. Clean-room n8n. The Neon DB already held a prior n8n instance's workflows + credentials, but its encryption key is unknown (can't decrypt). Ran a clean local n8n (own sqlite, fixed encryption key) and re-created the 5 credentials (Voyage / DeepSeek / Gemini / OpenAI / Neon-Postgres) from the API keys via `n8n import:credentials`.

B3. Backend Python 3.13 pins. Bumped vs `backend/requirements.txt`: `psycopg2-binary>=2.9.10`, `asyncpg>=0.30`, `sqlalchemy>=2.0.36` (2.0.30 crashes on 3.13), and `bcrypt 4.x` (passlib 1.7.4 fatally breaks on bcrypt 5).

B4. RAGAS judge venv. `ragas==0.1.10` is incompatible with modern langchain. Used a separate venv with era-correct pins: `langchain 0.1.20`, `langchain-community 0.0.38`, `langchain-core 0.1.53`, `langchain-openai 0.1.7`, `openai<2`, `datasets<3`.

B5. Backend wiring. `pipeline_client` reads `PIPELINE_URL` via raw `os.getenv` (not `.env`), so exported `PIPELINE_URL=http://localhost:5678/webhook/retrieve` and ran with `MOCK_N8N=false`. Schema created via `database/init.sql`; the 28 suite docs ingested directly into `document_chunks` for the `iisc-demo` tenant (512-word/50-overlap, Voyage `voyage-4-large`, `input_type=document`).

C. Evaluation harness fixes (`evaluation/run_eval.py`)

C1. RAGAS scoring crash. Scoring failed with *"This event loop is already running"* and *"LLM is not set"*. **Fix:** `nest_asyncio.apply()` + explicitly bind the judge LLM/embeddings (`gpt-4o-mini`, `text-embedding-3-small`) onto each metric.

C2. Run resilience. A single flaky query (502) aborted the whole 148-case run. Added per-case retry with backoff (5 attempts) and **skip-on-persistent-failure** so one bad case can't kill the run; raised the backend `PIPELINE_TIMEOUT_SECONDS 120→300` and added 1.5s pacing to avoid DeepSeek rate-limit windows.

C3. Output persistence + auth. Persist raw collected outputs immediately (re-score via `--from-outputs` without re-running 148 queries); switched to JWT bearer-token auth for the runner (the admin password hash kept reverting between runs).

3. Results by Application

Green = meets threshold, red = below. Context recall is a perfect 1.00 everywhere; precision (0.93 avg) is the more informative retrieval signal. Answer relevancy is the consistently weak metric.

Application	n	Faithf.	Ans.Rel.	Ctx.Prec.	Ctx.Rec.	Lat (ms)
bug_reporting	30	0.92	0.67	0.94	1.00	5,770
compliance_policy	8	0.86	0.75	0.89	1.00	11,270
customer_support	8	0.80	0.74	0.95	1.00	5,126
developer_documentation	7	0.86	0.63	0.98	1.00	9,336
education_assistant	8	0.83	0.83	1.00	1.00	19,810
employee_onboarding	8	0.86	0.64	0.95	1.00	5,937
equipment_maintenance	7	0.71	0.56	0.96	1.00	6,241
healthcare_administration	8	0.86	0.70	0.92	1.00	5,484
incident_response	7	0.83	0.67	0.93	1.00	10,849
it_helpdesk	8	1.00	0.73	0.94	1.00	5,336
kubernetes_troubleshooting	30	0.84	0.62	0.92	1.00	6,181
legal_document_navigation	8	1.00	0.73	0.94	1.00	5,133
sales_enablement	8	0.71	0.51	0.85	1.00	6,853

Reading the metrics. **Faithfulness** = answer claims supported by retrieved context. **Answer relevancy** = answer directly addresses the question (penalizes terse/partial answers). **Context precision** = retrieved chunks are relevant & well-ranked. **Context recall** = the needed evidence was retrieved. Thresholds: faithf. 0.85 · ans.rel. 0.80 · ctx.prec. 0.75 · ctx.rec. 0.80.

4. Caveats

- **3 cases skipped** on transient DeepSeek 502s under sustained load: DEVELOPER_DOCUMENTATION-008 , INCIDENT_RESPONSE-008 , EQUIPMENT_MAINTENANCE-008 . Infrastructure failures, not model quality — but a reliability finding: the DeepSeek-primary chain is not robust under load and the fallback path did not always catch it.
- Synthetic source docs are small (~1 chunk each), which makes context recall easy to achieve; precision and the generation metrics are the more meaningful signals here.
- Ingestion (M5) was bypassed (docs inserted directly) to isolate the M6 *retrieval* module under test.

5. Recommendations

- **Fold the four group-A fixes into the committed M6 workflow** (owner: M6) — without them the pipeline returns no citations and abstains on everything.
- **Raise answer relevancy** (the only failing gate): prompt the generator to answer in a complete, self-contained sentence rather than a bare fragment.
- **Harden the generation fallback** so DeepSeek rate-limit/5xx reliably routes to OpenAI instead of 502-ing.

6. Per-Case Results (Appendix – 145 cases)

ID	Application	Category	Ans?	Question	Faith	A.Rel	C.Prec	C.Rec	Lat
K8S-EVAL-001	kubernetes_troubleshooting	factoid	Y	Which sources should the Kubernetes troubleshooting knowledge base pri	—	0.76	1.00	—	5,630
K8S-EVAL-002	kubernetes_troubleshooting	procedural	Y	What fields belong in the initial triage packet for a Kubernetes incid	—	0.92	0.96	—	6,791
K8S-EVAL-003	kubernetes_troubleshooting	procedural	Y	What should be checked first when a pod stays Pending?	—	0.84	1.00	1.00	5,755
K8S-EVAL-004	kubernetes_troubleshooting	procedural	Y	How should CrashLoopBackOff be investigated?	—	—	1.00	—	5,975
K8S-EVAL-005	kubernetes_troubleshooting	factoid	Y	What does ImagePullBackOff usually indicate?	1.00	—	1.00	1.00	5,014
K8S-EVAL-006	kubernetes_troubleshooting	factoid	Y	What usually causes CreateContainerConfigError?	1.00	0.91	1.00	1.00	5,628
K8S-EVAL-007	kubernetes_troubleshooting	procedural	Y	Which signals should be inspected for node pressure evictions?	1.00	0.69	0.81	1.00	6,783
K8S-EVAL-008	kubernetes_troubleshooting	procedural	Y	What is the safe starting point for control-plane troubleshooting?	1.00	0.69	1.00	1.00	6,046
K8S-EVAL-009	kubernetes_troubleshooting	procedural	Y	A Service has no endpoints. What comparisons should be made?	1.00	0.32	0.59	1.00	5,047
K8S-EVAL-010	kubernetes_troubleshooting	procedural	Y	How should in-cluster DNS resolution failure be debugged?	1.00	0.66	1.00	1.00	5,363
K8S-EVAL-011	kubernetes_troubleshooting	procedural	Y	What must be checked when NetworkPolicy unexpectedly blocks traffic?	1.00	0.54	1.00	1.00	5,259
K8S-EVAL-012	kubernetes_troubleshooting	procedural	Y	How should an Ingress outage be decomposed?	1.00	0.55	0.68	—	5,854
K8S-EVAL-013	kubernetes_troubleshooting	factoid	Y	Why can a PersistentVolumeClaim remain Pending?	1.00	0.94	1.00	—	5,721
K8S-EVAL-014	kubernetes_troubleshooting	procedural	Y	What evidence should be gathered for volume attach or detach failures?	—	0.66	0.81	—	5,193
K8S-EVAL-015	kubernetes_troubleshooting	reasoning	Y	Why might an HPA not change replica count?	1.00	0.93	1.00	—	6,996
K8S-EVAL-016	kubernetes_troubleshooting	safety	Y	What production debugging actions should be avoided?	1.00	0.53	0.65	1.00	5,839
K8S-EVAL-017	kubernetes_troubleshooting	procedural	Y	On EKS, what should be checked before recreating nodes for pod network	—	—	1.00	1.00	4,863
K8S-EVAL-018	kubernetes_troubleshooting	procedural	Y	What should be verified before changing RBAC for an EKS authentication	1.00	0.90	0.78	1.00	5,440
K8S-EVAL-019	kubernetes_troubleshooting	reasoning	Y	How should GKE Autopilot scheduling denials be handled?	1.00	0.58	1.00	1.00	5,145
K8S-EVAL-020	kubernetes_troubleshooting	multi_hop	Y	How can a GKE logging anomaly cascade into node pressure?	1.00	—	1.00	—	5,066
K8S-EVAL-021	kubernetes_troubleshooting	factoid	Y	On AKS with Azure CNI, what can cause pod connectivity problems?	1.00	0.94	1.00	1.00	5,835

K8S-EVAL-022	kubernetes_troubleshooting	procedural	Y	What should be done after AKS identity or role-assignment changes trig	1.00	0.71	0.83	1.00	5,802
K8S-EVAL-023	kubernetes_troubleshooting	reasoning	Y	What pattern should the evaluator reward in Kubernetes post-mortem ans	—	0.62	0.94	1.00	5,900
K8S-EVAL-024	kubernetes_troubleshooting	procedural	Y	When should a Kubernetes issue be escalated to platform or cloud-provi	—	0.79	1.00	—	6,286
K8S-EVAL-025	kubernetes_troubleshooting	multi_hop	Y	A pod is Pending and its PVC is also Pending. Which root causes should	0.00	0.00	1.00	—	9,266
K8S-EVAL-026	kubernetes_troubleshooting	multi_hop	Y	A Service has no endpoints after a rollout and users see Ingress 503s.	—	0.39	0.86	1.00	7,206
K8S-EVAL-027	kubernetes_troubleshooting	multi_hop	Y	A CrashLoopBackOff started after a ConfigMap change. Which evidence co	0.00	0.00	1.00	—	12,286
K8S-EVAL-028	kubernetes_troubleshooting	safety	Y	Is it safe to collect a full environment dump from a production pod to	0.00	0.00	0.90	—	7,278
K8S-EVAL-029	kubernetes_troubleshooting	unanswerable	N	Which Helm chart version fixes the documented CoreDNS timeout issue?	—	—	—	—	6,420
K8S-EVAL-030	kubernetes_troubleshooting	unanswerable	N	What is the monthly price of a larger managed Kubernetes node pool?	—	—	—	—	5,758
BUG-EVAL-001	bug_reporting	factoid	Y	What information is mandatory in a new bug report?	1.00	—	0.66	—	5,245
BUG-EVAL-002	bug_reporting	factoid	Y	Which situations qualify as severity S1?	—	0.70	0.96	1.00	5,573
BUG-EVAL-003	bug_reporting	factoid	Y	How quickly must an S1 issue be acknowledged, and what else must happe	1.00	0.78	0.83	1.00	4,909
BUG-EVAL-004	bug_reporting	factoid	Y	What is the acknowledgement target for an S2 report?	1.00	0.72	0.73	1.00	4,583
BUG-EVAL-005	bug_reporting	reasoning	Y	How should a cosmetic documentation typo be classified and acknowledge	—	0.85	1.00	1.00	5,786
BUG-EVAL-006	bug_reporting	procedural	Y	What steps should a triager take when a report is a duplicate?	—	0.56	1.00	1.00	6,154
BUG-EVAL-007	bug_reporting	factoid	Y	Where should a suspected vulnerability be filed?	1.00	0.80	0.70	1.00	5,365
BUG-EVAL-008	bug_reporting	safety	Y	Can an access token be pasted into a public issue for debugging?	—	0.00	0.78	1.00	8,394
BUG-EVAL-009	bug_reporting	factoid	Y	When do repeated support reports trigger the incident process?	0.50	0.70	0.81	1.00	5,567
BUG-EVAL-010	bug_reporting	reasoning	Y	An S2 defect affects 30 percent of active tenants. Should it enter the	—	0.89	0.96	1.00	6,548
BUG-EVAL-011	bug_reporting	factoid	Y	Which devices and app versions are affected by BUG-MOB-101?	1.00	0.92	1.00	—	5,409
BUG-EVAL-012	bug_reporting	factoid	Y	What is the workaround for the Android login loop?	1.00	0.77	1.00	1.00	4,740
BUG-EVAL-013	bug_reporting	factoid	Y	Which version permanently fixes BUG-MOB-101?	1.00	0.44	1.00	1.00	5,642
BUG-EVAL-014	bug_reporting	factoid	Y	What triggers BUG-MOB-102?	0.67	0.93	1.00	1.00	5,417
BUG-EVAL-015	bug_reporting	factoid	Y	How can a user avoid the attachment crash?	1.00	0.46	1.00	1.00	5,739

BUG-EVAL-016	bug_reporting	factoid	Y	What should an iOS 17 user do when notifications remain missing after	0.50	0.49	1.00	1.00	4,846
BUG-EVAL-017	bug_reporting	factoid	Y	Which version fixes the missing iOS notification issue?	1.00	0.48	1.00	1.00	4,927
BUG-EVAL-018	bug_reporting	procedural	Y	What evidence should be collected for a mobile crash report?	1.00	0.88	0.96	—	5,334
BUG-EVAL-019	bug_reporting	multi_hop	Y	Compare the workarounds for BUG-MOB-101 and BUG-MOB-102.	—	—	1.00	—	5,675
BUG-EVAL-020	bug_reporting	unanswerable	N	What is the official password reset policy for the mobile app?	—	—	—	—	5,647
BUG-EVAL-021	bug_reporting	factoid	Y	Why can HTTP 429 responses appear after an access-token rotation?	1.00	0.96	1.00	1.00	5,603
BUG-EVAL-022	bug_reporting	factoid	Y	What is the workaround for BUG-API-201?	1.00	0.46	1.00	1.00	4,838
BUG-EVAL-023	bug_reporting	factoid	Y	Which API release contains the server-side fix for BUG-API-201?	1.00	—	1.00	1.00	5,082
BUG-EVAL-024	bug_reporting	factoid	Y	Why might the same webhook event be delivered twice?	—	0.59	1.00	1.00	5,273
BUG-EVAL-025	bug_reporting	procedural	Y	How should consumers prevent duplicate webhook processing?	1.00	0.38	1.00	1.00	8,127
BUG-EVAL-026	bug_reporting	factoid	Y	Does duplicate webhook delivery mean the platform stored duplicate dat	1.00	0.80	1.00	1.00	5,851
BUG-EVAL-027	bug_reporting	factoid	Y	What are the documented causes of webhook signature validation failure	1.00	0.93	1.00	—	6,845
BUG-EVAL-028	bug_reporting	procedural	Y	What details belong in an API bug report, and what sensitive value mus	0.78	—	0.87	1.00	7,913
BUG-EVAL-029	bug_reporting	unanswerable	N	Which GraphQL mutations are affected by the current known issues?	—	—	—	—	5,664
BUG-EVAL-030	bug_reporting	unanswerable	N	What is the price of the enterprise API plan?	—	—	—	—	6,416
IT_HELPDESK-001	it_helpdesk	factoid	Y	How long is a password reset link valid?	1.00	0.94	1.00	1.00	4,411
IT_HELPDESK-002	it_helpdesk	procedural	Y	What should I do for VPN error 720?	—	0.98	1.00	1.00	5,337
IT_HELPDESK-003	it_helpdesk	procedural	Y	How is MFA recovered when the old phone is unavailable?	1.00	0.39	1.00	—	5,349
IT_HELPDESK-004	it_helpdesk	factoid	Y	Can employees install standard software without administrator rights?	1.00	0.80	1.00	1.00	5,245
IT_HELPDESK-005	it_helpdesk	reasoning	Y	What priority applies when laptop encryption status is missing?	1.00	0.76	0.64	1.00	5,413
IT_HELPDESK-006	it_helpdesk	procedural	Y	What evidence belongs in a helpdesk ticket?	1.00	0.83	0.95	—	5,160
IT_HELPDESK-007	it_helpdesk	multi_hop	Y	A locked user also lost the old MFA phone. What timing constraints app	1.00	0.44	1.00	1.00	6,716
IT_HELPDESK-008	it_helpdesk	unanswerable	N	What is the company Wi-Fi password?	—	—	—	—	5,057
CUSTOMER_SUPPORT-001	customer_support	factoid	Y	How long is the NovaHub hardware warranty?	1.00	0.86	1.00	1.00	5,583

CUSTOMER_SUPPORT-002	customer_support	factoid	Y	Can accidental damage be claimed under warranty?	1.00	0.48	1.00	1.00	4,741
CUSTOMER_SUPPORT-003	customer_support	procedural	Y	How are opened defective products returned?	1.00	0.84	1.00	1.00	4,738
CUSTOMER_SUPPORT-004	customer_support	factoid	Y	How quickly does an approved replacement ship?	1.00	0.65	1.00	1.00	4,938
CUSTOMER_SUPPORT-005	customer_support	procedural	Y	How do I factory-reset NovaHub?	—	0.90	1.00	1.00	4,487
CUSTOMER_SUPPORT-006	customer_support	procedural	Y	What should be tried when NovaHub is offline?	—	0.87	1.00	—	5,168
CUSTOMER_SUPPORT-007	customer_support	multi_hop	Y	The unit overheats after repeated resets. What should support do?	0.00	0.61	0.66	1.00	6,070
CUSTOMER_SUPPORT-008	customer_support	unanswerable	N	Does NovaHub support satellite internet?	—	—	—	—	5,284
EMPLOYEE_ONBOARDING-001	employee_onboarding	factoid	Y	What time should a new employee report on the first day?	1.00	0.98	1.00	1.00	5,059
EMPLOYEE_ONBOARDING-002	employee_onboarding	procedural	Y	What happens before system access is issued?	1.00	0.27	1.00	1.00	5,391
EMPLOYEE_ONBOARDING-003	employee_onboarding	factoid	Y	When must mandatory training be completed?	1.00	0.89	0.67	1.00	5,337
EMPLOYEE_ONBOARDING-004	employee_onboarding	factoid	Y	How quickly should courier damage be reported?	1.00	0.58	1.00	1.00	5,126
EMPLOYEE_ONBOARDING-005	employee_onboarding	factoid	Y	When are probation check-ins held?	1.00	0.97	1.00	1.00	5,279
EMPLOYEE_ONBOARDING-006	employee_onboarding	procedural	Y	How is planned leave requested?	1.00	0.76	1.00	1.00	5,171
EMPLOYEE_ONBOARDING-007	employee_onboarding	reasoning	Y	Can onboarding documents be sent to a personal email to finish trainin	0.00	0.00	1.00	1.00	9,661
EMPLOYEE_ONBOARDING-008	employee_onboarding	unanswerable	N	What is the employee cafeteria menu?	—	—	—	—	6,475
DEVELOPER_DOCUMENTATION-001	developer_documentation	factoid	Y	How are Payments API requests authenticated?	1.00	0.83	0.83	1.00	5,260
DEVELOPER_DOCUMENTATION-002	developer_documentation	factoid	Y	What header is required when creating a payment?	1.00	0.57	1.00	1.00	4,859
DEVELOPER_DOCUMENTATION-003	developer_documentation	factoid	Y	How long may an idempotency key be reused?	1.00	0.79	1.00	1.00	5,859
DEVELOPER_DOCUMENTATION-004	developer_documentation	factoid	Y	What does HTTP 409 mean?	1.00	0.97	1.00	—	5,517
DEVELOPER_DOCUMENTATION-005	developer_documentation	factoid	Y	Which Python versions are officially supported?	1.00	0.68	1.00	1.00	5,073
DEVELOPER_DOCUMENTATION-006	developer_documentation	procedural	Y	How should a webhook signature be verified?	1.00	0.60	1.00	1.00	18,063
DEVELOPER_DOCUMENTATION-007	developer_documentation	multi_hop	Y	How should a Node.js client retry a rate-limited payment creation safe	0.00	0.00	1.00	1.00	20,718
INCIDENT_RESPONSE-001	incident_response	factoid	Y	When should an incident be declared?	1.00	0.82	0.98	1.00	35,900
INCIDENT_RESPONSE-002	incident_response	factoid	Y	Who owns incident decisions and timeline updates?	1.00	0.85	0.77	1.00	5,098
INCIDENT_RESPONSE-003	incident_response	procedural	Y	What must happen in the first 15 minutes?	1.00	0.60	0.92	1.00	5,013
INCIDENT_RESPONSE-004	incident_response	factoid	Y	How often are SEV-1 customers updated?	1.00	0.81	1.00	1.00	4,909
INCIDENT_RESPONSE-005	incident_response	reasoning	Y	When is rollback preferred?	—	0.86	1.00	—	5,606
INCIDENT_RESPONSE-006	incident_response	procedural	Y	How should original incident evidence be handled?	1.00	0.74	0.91	1.00	6,464
INCIDENT_RESPONSE-007	incident_response	multi_hop	Y	A SEV-1 follows a recent deployment. What	0.00	0.00	0.97	1.00	12,950

				immediate and follow-up acti					
COMPLIANCE_POLICY-001	compliance_policy	factoid	Y	How are payment records classified?	1.00	0.95	0.64	1.00	34,519
COMPLIANCE_POLICY-002	compliance_policy	factoid	Y	How long are support tickets retained?	1.00	1.00	0.61	1.00	16,109
COMPLIANCE_POLICY-003	compliance_policy	factoid	Y	How often is Restricted-data access reviewed?	1.00	0.94	1.00	1.00	8,327
COMPLIANCE_POLICY-004	compliance_policy	procedural	Y	What is required before a vendor processes Restricted data?	1.00	—	1.00	1.00	5,048
COMPLIANCE_POLICY-005	compliance_policy	factoid	Y	How quickly must suspected Restricted-data exposure be reported?	1.00	0.80	1.00	1.00	5,321
COMPLIANCE_POLICY-006	compliance_policy	reasoning	Y	Why is an untimestamped screenshot insufficient evidence?	1.00	0.80	0.94	—	5,826
COMPLIANCE_POLICY-007	compliance_policy	multi_hop	Y	A vendor stores payment records. Which classification and review requi	0.00	0.00	1.00	1.00	8,832
COMPLIANCE_POLICY-008	compliance_policy	unanswerable	N	What fine applies for a late access review?	—	—	—	—	6,176
EDUCATION_ASSISTANT-001	education_assistant	factoid	Y	How much is the course project worth?	1.00	0.69	1.00	1.00	9,456
EDUCATION_ASSISTANT-002	education_assistant	factoid	Y	What score is required to pass?	1.00	0.76	1.00	1.00	25,528
EDUCATION_ASSISTANT-003	education_assistant	reasoning	Y	What happens to an assignment submitted two days late?	0.33	0.73	1.00	—	23,385
EDUCATION_ASSISTANT-004	education_assistant	factoid	Y	When is the project checkpoint?	1.00	0.93	1.00	1.00	22,408
EDUCATION_ASSISTANT-005	education_assistant	factoid	Y	How many GPU hours may a student reserve per day?	1.00	0.98	1.00	1.00	17,077
EDUCATION_ASSISTANT-006	education_assistant	procedural	Y	What makes an experiment submission reproducible?	—	0.69	1.00	1.00	22,883
EDUCATION_ASSISTANT-007	education_assistant	multi_hop	Y	Can a team reuse public training code in its project?	0.67	1.00	1.00	1.00	27,160
EDUCATION_ASSISTANT-008	education_assistant	unanswerable	N	Which textbook chapter is assigned for week 7?	—	—	—	—	10,580
HEALTHCARE_ADMINISTRATION-001	healthcare_administration	factoid	Y	When can a routine appointment be rescheduled without charge?	1.00	0.91	1.00	1.00	4,838
HEALTHCARE_ADMINISTRATION-002	healthcare_administration	procedural	Y	How does a patient request medical records?	1.00	0.78	1.00	1.00	6,181
HEALTHCARE_ADMINISTRATION-003	healthcare_administration	factoid	Y	How quickly are records requests acknowledged?	1.00	1.00	1.00	1.00	4,653
HEALTHCARE_ADMINISTRATION-004	healthcare_administration	safety	Y	May a minor child interpret a consent discussion?	1.00	0.85	1.00	—	4,852
HEALTHCARE_ADMINISTRATION-005	healthcare_administration	procedural	Y	What details are needed for a billing dispute?	1.00	0.90	0.83	1.00	5,082
HEALTHCARE_ADMINISTRATION-006	healthcare_administration	procedural	Y	What should staff do after records are sent to the wrong recipient?	1.00	0.48	0.61	1.00	5,562
HEALTHCARE_ADMINISTRATION-007	healthcare_administration	reasoning	Y	Can administrative staff explain a laboratory result while handling a	0.00	0.00	1.00	1.00	6,282
HEALTHCARE_ADMINISTRATION-008	healthcare_administration	unanswerable	N	Which medication should a patient take for a fever?	—	—	—	—	6,422
LEGAL_DOCUMENT_NAVIGATION-001	legal_document_navigation	factoid	Y	When does the agreement begin and how long is the initial term?	1.00	0.88	1.00	1.00	5,092
LEGAL_DOCUMENT_NAVIGATION-002	legal_document_navigation	factoid	Y	How much renewal notice is required?	1.00	0.55	1.00	1.00	5,056

LEGAL_DOCUMENT_NAVIGATION-003	legal_document_navigation	factoid	Y	When are invoices due?	1.00	1.00	1.00	1.00	5,027
LEGAL_DOCUMENT_NAVIGATION-004	legal_document_navigation	factoid	Y	How long does confidentiality survive termination?	1.00	0.67	1.00	1.00	5,067
LEGAL_DOCUMENT_NAVIGATION-005	legal_document_navigation	factoid	Y	When is an email notice effective?	1.00	0.91	1.00	1.00	4,849
LEGAL_DOCUMENT_NAVIGATION-006	legal_document_navigation	procedural	Y	What cure period applies to a material breach?	1.00	0.47	1.00	1.00	4,871
LEGAL_DOCUMENT_NAVIGATION-007	legal_document_navigation	multi_hop	Y	How should a party terminate for uncured material breach?	1.00	0.66	0.60	1.00	4,835
LEGAL_DOCUMENT_NAVIGATION-008	legal_document_navigation	unanswerable	N	Will a court enforce the renewal clause?	—	—	—	—	6,271
EQUIPMENT_MAINTENANCE-001	equipment_maintenance	procedural	Y	What must be inspected before each shift?	1.00	0.91	1.00	—	4,638
EQUIPMENT_MAINTENANCE-002	equipment_maintenance	safety	Y	When must the press not be operated?	0.00	0.00	1.00	1.00	8,698
EQUIPMENT_MAINTENANCE-003	equipment_maintenance	factoid	Y	How often are the guide rails lubricated?	1.00	0.98	1.00	1.00	5,054
EQUIPMENT_MAINTENANCE-004	equipment_maintenance	procedural	Y	What should be checked before replacing a pump for pressure drift?	1.00	0.77	0.83	1.00	5,855
EQUIPMENT_MAINTENANCE-005	equipment_maintenance	safety	Y	What lockout steps are required inside the guarded area?	1.00	0.74	1.00	1.00	5,728
EQUIPMENT_MAINTENANCE-006	equipment_maintenance	factoid	Y	At what temperature must the press stop?	1.00	0.52	1.00	1.00	4,540
EQUIPMENT_MAINTENANCE-007	equipment_maintenance	multi_hop	Y	A leaking hose has exposed reinforcement and the fluid is 78 degrees.	0.00	0.00	0.92	1.00	9,172
SALES_ENABLEMENT-001	sales_enablement	factoid	Y	How many users does CloudDesk Standard support?	1.00	1.00	1.00	1.00	14,747
SALES_ENABLEMENT-002	sales_enablement	factoid	Y	Which plan includes SSO and audit export?	1.00	0.35	1.00	1.00	5,580
SALES_ENABLEMENT-003	sales_enablement	factoid	Y	How long is trial data retained after expiration?	0.00	0.00	0.50	1.00	6,457
SALES_ENABLEMENT-004	sales_enablement	safety	Y	May sales promise that CloudDesk guarantees compliance?	1.00	0.89	0.87	1.00	4,640
SALES_ENABLEMENT-005	sales_enablement	reasoning	Y	Who approves a 15 percent discount?	1.00	0.67	1.00	1.00	5,015
SALES_ENABLEMENT-006	sales_enablement	procedural	Y	What should a seller do when the security brief lacks an answer?	1.00	0.62	0.83	—	5,074
SALES_ENABLEMENT-007	sales_enablement	multi_hop	Y	A regulated customer needs SSO and asks for a 15 percent discount. Wha	0.00	0.00	0.77	1.00	7,370
SALES_ENABLEMENT-008	sales_enablement	unanswerable	N	What exact price does a competitor charge?	—	—	—	—	5,941

Blank metric cells = unanswerable cases (not scored by RAGAS; evaluated instead via abstention). Full machine-readable data: [application_suite_eval_latest.json, csv, html](#) .